Causality Assessment: Drug-Reaction Relationship

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1. Objective

To evaluate the relationship between a drug and an adverse reaction using standard causality assessment methods (WHO-UMC and Naranjo), and to classify reactions as Certain, Probable or Possible. Four hypothetical cases are assessed. One extra case shows an Unlikely result, so that the full scale is covered.

2. Why causality assessment matters

Causality assessment judges how likely it is that a drug caused an event, since a suspected reaction is not proof. It guides regulatory action, labelling changes, signal detection and clinical decisions. No method is perfect, so a structured scale makes judgements consistent and open to review.

3. The WHO-UMC system

This is the qualitative scale most widely used in pharmacovigilance, including by PvPI.

Category	Criteria
Certain	Plausible time relationship. Cannot be explained by disease or other drugs. Response to withdrawal is plausible. Event is pharmacologically or phenomenologically definitive. Rechallenge satisfactory, if necessary.
Probable / Likely	Reasonable time relationship. Unlikely to be due to disease or other drugs. Withdrawal response is clinically reasonable. Rechallenge not required.
Possible	Reasonable time relationship. Could also be explained by disease or other drugs. Withdrawal information may be lacking or unclear.
Unlikely	Time relationship improbable (but not impossible). Disease or other drugs provide plausible explanations.
Conditional / Unclassified	More data are essential for a proper assessment, or are under examination.
Unassessable / Unclassifiable	Information is insufficient or contradictory and cannot be supplemented or verified.

4. The Naranjo algorithm

A quantitative scale of ten questions. Each is answered Yes, No or Do not know, and the scores are added.

No.	Question	Yes	No	Do not know
1	Are there previous conclusive reports on this reaction?	+1	0	0
2	Did the event appear after the suspected drug was given?	+2	-1	0
3	Did the reaction improve when the drug was stopped or a specific antagonist given?	+1	0	0
4	Did the reaction reappear when the drug was re-administered?	+2	-1	0
5	Are there alternative causes that could on their own have caused the reaction?	-1	+2	0
6	Did the reaction reappear when a placebo was given?	-1	+1	0
7	Was the drug detected in blood or other fluids in toxic concentrations?	+1	0	0
8	Was the reaction more severe when the dose was increased, or less	+1	0	0

No.	Question	Yes	No	Do not know
	severe when decreased?			
9	Did the patient have a similar reaction to the same or similar drugs before?	+1	0	0
10	Was the event confirmed by objective evidence?	+1	0	0

Interpretation: Definite 9 or more. Probable 5 to 8. Possible 1 to 4. Doubtful 0 or less.

5. Case studies

All cases are hypothetical and written for training. Laboratory values are illustrative.

Case 1: Fixed drug eruption with co-trimoxazole (Certain)

Scenario: A 42-year-old woman with a urinary tract infection took co-trimoxazole (800/160 mg) twice daily. Three hours after the first dose she developed a sharply demarcated, round, itchy, violaceous plaque of about 4 cm on the right forearm, which later blistered. She had no other new medicines and no systemic symptoms. Co-trimoxazole was stopped and a topical corticosteroid was applied. The lesion healed in 10 days, leaving a residual brown patch. Skin biopsy showed interface dermatitis with necrotic keratinocytes and pigment incontinence, consistent with fixed drug eruption. After healing, a supervised oral provocation test with a quarter dose reproduced a lesion at the same site within hours.

Analysis:

- Time relationship: plausible (hours after the dose, typical for fixed drug eruption).
- Alternative causes: none identified, no other new drug.
- Dechallenge: positive. Rechallenge: positive, done under medical supervision.
- Event definitive: fixed drug eruption is a recognised drug-specific phenomenon, and sulfonamides are common causes.

Naranjo score 9 (Definite): previous reports +1, event after drug +2, improved on stopping +1, reappeared on rechallenge +2, no alternative cause +2, objective evidence (biopsy) +1.

WHO-UMC: Certain. All criteria are met, including a satisfactory rechallenge.

Seriousness and severity: Non-serious. Hartwig level 2.

Case 2: Tendinopathy with ciprofloxacin (Probable)

Scenario: A 63-year-old man received ciprofloxacin 500 mg twice daily for 7 days for a urinary tract infection. On day 5 he developed pain, swelling and stiffness in the right Achilles tendon and could not walk comfortably. He had no trauma, no recent unusual exertion and no corticosteroid use. Ultrasound showed tendon thickening with hypoechoic areas, consistent with tendinopathy. Ciprofloxacin was stopped on day 5, with rest and analgesia, and symptoms improved over three weeks. The drug was not re-administered.

Analysis:

- Time relationship: reasonable (fluoroquinolone tendinopathy usually appears within days to weeks).
- Alternative causes: none found on history and examination.
- Dechallenge: positive. Rechallenge: not done, and not appropriate.

Naranjo score 7 (Probable): previous reports +1, event after drug +2, improved on stopping +1, no alternative cause +2, objective evidence (ultrasound) +1.

WHO-UMC: Probable / Likely. Rechallenge is not required for this category, and would be unsafe.

Seriousness and severity: Non-serious but significant disability for several weeks. Hartwig level 3 (drug stopped, treatment required). Age over 60 is a recognised risk factor.

Case 3: Hepatotoxicity during anti-tuberculosis therapy (Possible)

Scenario: A 35-year-old man with pulmonary tuberculosis started the fixed-dose combination of isoniazid, rifampicin, pyrazinamide and ethambutol. At week 6 he developed nausea, dark urine and jaundice. ALT was 480 U/L (more than 10 times the upper limit of normal) and total bilirubin 4.2 mg/dL. Hepatitis A, B and C serology was negative, he had no alcohol use, and abdominal ultrasound was normal. All hepatotoxic drugs were stopped. Liver tests returned to normal within six weeks. A supervised, sequential reintroduction was planned but not yet done.

Analysis:

- Time relationship: reasonable (usually within the first two months).
- Alternative causes: viral hepatitis and alcohol excluded, but three of the four drugs are individually hepatotoxic, so the culprit cannot be separated.
- Dechallenge: positive for the regimen as a whole, not for any single drug.
- Rechallenge: not yet done.

Naranjo score 4 for isoniazid (Possible): previous reports +1, event after drug +2, improved on stopping +1, alternative causes (other hepatotoxic drugs) -1, objective evidence (ALT, bilirubin) +1.

WHO-UMC: Possible, for each of isoniazid, rifampicin and pyrazinamide. The category can be upgraded once sequential rechallenge identifies the responsible drug.

Seriousness and severity: Serious (hospitalisation, medically important). Hartwig level 4(b). Management follows national TB programme guidance for drug-induced liver injury.

Case 4 (extra): Rash with pantoprazole (Unlikely)

Scenario: A 30-year-old woman started pantoprazole 40 mg daily for gastro-oesophageal reflux. She reported an itchy rash on the hands and waist that she noticed two days before the first dose. Skin examination showed burrows and excoriations typical of scabies. The rash continued after pantoprazole was stopped and settled only after scabicide treatment.

Analysis: The event began before the drug was given, did not improve after withdrawal, and has a clear alternative cause.

Naranjo score -1 (Doubtful): previous reports +1, event after drug -1, no improvement on stopping 0, alternative cause -1.

WHO-UMC: Unlikely.

Action: Pantoprazole is not the cause. Continue it if needed, treat scabies and treat close contacts. It is still worth documenting, since an "Unlikely" result is a valid outcome.

6. Summary of the assessments

Case	Suspected drug	Reaction	Naranjo	WHO-UMC	Key deciding factor
1	Co-trimoxazole	Fixed drug eruption	9 (Definite)	Certain	Positive rechallenge
2	Ciprofloxacin	Tendinopathy	7 (Probable)	Probable	Positive dechallenge, no alternative cause
3	Isoniazid (with rifampicin, pyrazinamide)	Hepatotoxicity	4 (Possible)	Possible	Several hepatotoxic drugs together
4	Pantoprazole	Rash	-1 (Doubtful)	Unlikely	Rash began before the drug

7. Comparing the two scales

Feature	WHO-UMC	Naranjo
Type	Qualitative, category based	Quantitative, score based
Strength	Quick, flexible, accepts clinical judgement	Structured, reproducible, easy to teach

Feature	WHO-UMC	Naranjo
Limitation	More subjective, assessors may differ	Rigid, penalises cases with unavoidable missing data, and needs rechallenge for the top score
Best use	Routine spontaneous reports (used by PvPI)	Research, training, clinical case reports

Both scales can disagree slightly. Where they do, the reasoning should be stated and the more conservative category reported.

8. Key learning points

- A positive rechallenge is the strongest single piece of evidence, but it is rarely ethical for serious reactions.
- The time relationship must be checked first. If the event came before the drug, causality is unlikely.
- Concomitant drugs and disease lower the category. Write them down.
- Dechallenge is more useful for drugs with short half-lives and reversible reactions.
- "Possible" is not a weak result. It is an honest one, and the case can be upgraded as more information arrives.
- Always report the reasoning with the category, not the label alone.

9. Reusable assessment worksheet

Step	Record
1. Event	Reaction term, onset date, severity, outcome
2. Drugs	Suspect and concomitant drugs with doses and dates
3. Timeline	Drug start, onset, dechallenge, rechallenge
4. Other causes	Disease, infection, other drugs, lab error
5. Known profile	Is the reaction listed in the label or literature?
6. Naranjo	Score each of the ten questions and total
7. WHO-UMC	Choose the category and justify it in one or two sentences
8. Follow-up	Further data needed, report to PvPI, patient advice

10. Assumptions and limitations

The cases are hypothetical, and findings such as laboratory values and test results are illustrative. Causality ratings depend on the information available and can reasonably differ between assessors. A real case should be assessed with the full clinical record.

11. References

- WHO-UMC. The use of the WHO-UMC system for standardised case causality assessment.
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- Indian Pharmacopoeia Commission. Pharmacovigilance Programme of India guidance documents.
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